Modeling microRNA Repressive Pressure on the Hallmark Programs of Cancer

A conceptual and computational framework, applied to breast cancer

Abstract

A microRNA (miRNA) represses many target genes, and a gene is repressed by many miRNAs, so the biologically meaningful object is not a single binding event but a **field of repression** acting on a cell. We formalize this field as *pressure* — an evidence-weighted, expression-scaled potential for repression — and separate it cleanly from *coupling*, an observed, confounder-adjusted anti-correlation that asks whether the pressure is actually **realized** in a tumour. We model both quantities at five nested resolutions (the whole program universe, a single program, a miRNA's target set, a gene, and a single interaction), across the healthy→tumour trajectory, and we test realization at the messenger-RNA layer, the protein layer, and in independent patient cohorts. The framework is substrate-agnostic; here it is implemented on breast cancer using the fifty MSigDB Hallmark programs as the gene universe. This document presents the concepts, the formulas with their rationale, and the validation logic.

> **Central idea.** Pressure is a *potential* (what the wiring and abundance could deliver); coupling is *realization* (what is observed after confounders are removed); neither is causation (which requires a perturbation assay). Keeping these three rungs distinct — and joining potential with realization rather than conflating them — is the framework's organizing principle.

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1. The modeling problem and stance

1.1 What "pressure" is, and why it is a potential

A miRNA loaded into the RISC/Argonaute complex base-pairs a target mRNA and represses it (predominantly by destabilizing the message, with a smaller translational-block component). Because the regulatory graph is densely many-to-many, we quantify the aggregate downward force the miRNA layer places on a gene — its **pressure** — built from three ingredients:

- **which interactions exist** (curated and predicted edges),
- **how strong each interaction is** (assay evidence),
- **how abundant the miRNA is** (expression).

Pressure is deliberately a *potential* quantity. It does not, by itself, prove that repression occurred in a given tumour. That harder question is answered separately by **coupling**: a confounder-adjusted anti-correlation between the miRNA *dose* (its abundance at single-regulator resolution, or the aggregate pressure where regulators are pooled — §6) and its target's expression. The decision to hold these two as distinct measurements, and then to report their **join**, organizes everything that follows.

1.2 Why programs, why breast cancer

The fifty Hallmark gene sets are a curated, largely non-redundant atlas of the coherent biological *programs* of cancer (proliferation, epithelial–mesenchymal transition, hypoxia, apoptosis, p53, interferon response, and so on). Using them as the gene universe makes every result interpretable as "pressure on a named program," and lets pressure be rolled up and down a biological hierarchy.

Breast cancer is the implementation substrate because it offers the deepest matched multi-omics — miRNA, messenger RNA, copy number, DNA methylation, chromatin accessibility, and two independent mass-spectrometry proteomes — together with well-defined molecular subtypes and an independent replication cohort. Nothing in the pressure, coupling, or architecture machinery is breast-specific; only the program-polarity prior and one or two confounders are cancer-flavoured.

1.3 Design principles

These recur throughout and are stated once.

1. **Static evidence × dynamic abundance.** The interaction weight encodes curated and sequence evidence, which does not change between samples; *all* sample- and state-to-state variation enters through miRNA expression. This separates "what is biologically possible" from "what this tumour is expressing."
2. **The prior is non-circular.** Interaction weights, the evidence filter, the program-polarity prior, and gene-role annotations never use the patient expression data they are later tested against, so they cannot leak into the findings.
3. **Adjusted and raw, gated and ungated, always.** Every coupling carries both its raw and its confounder-adjusted value; every availability-gated pressure carries its ungated twin. Differences are read as sensitivity, not as the headline.
4. **Honest rungs of evidence.** Potential is not realization is not causation; each claim is labelled by the strength of support behind it.

1.4 Data, cohort, and stratification

All quantities are computed on **TCGA breast cancer (TCGA-BRCA)**: primary tumours with matched small-RNA sequencing and messenger-RNA sequencing, plus copy number, DNA methylation, and chromatin accessibility where assayed; the 104 normal-adjacent samples provide the intermediate tissue state (§8). The PAM50 **Normal-like** class (36 tumours; normal-tissue admixture) is **set aside cohort-wide**, leaving **1,065 retained tumours**; the realized-coupling tests run on the **~1,041** with complete miRNA + mRNA + covariate data. Over this cohort the interaction universe is **721 miRNA arms, 5,108 curated edges, and 1,410 target genes** drawn from the **50 MSigDB Hallmark programs** (4,384 genes in their union).

Tumours are stratified throughout by the **PAM50 molecular subtypes**, the field-standard breast taxonomy — and most relative quantities are *recomputed within each subtype*, not sliced from cohort numbers (§2):

PAM50 subtype	n (retained)	Character
Luminal A	507	hormone-receptor-positive, low-proliferation
Luminal B	264	hormone-receptor-positive, higher-proliferation
HER2-enriched	93	HER2-driven
Basal-like	197	largely triple-negative, high-proliferation
Normal-like	0 (36 set aside)	low-alteration, normal-tissue-like — **set aside cohort-wide** (normal-tissue admixture); excluded from the realized analyses

The external and reference datasets the framework draws on:

Source	Role
TCGA-BRCA	primary cohort — miRNA, mRNA, copy number, methylation, accessibility, subtype
GTEx (breast, 346 samples)	true-healthy miRNA reference for the acquired axis (§8)
DIANA-miTED	independent healthy miRNA atlas, used to repair healthy-reference coverage

Source	Role
CPTAC breast proteomes — a same-patient set (n = 105) and an *independent* cohort (n about 101)	protein-layer validation (§7)
Buffa 2011 (210 paired primary breast tumours, no TCGA overlap)	independent-cohort replication of coupling (§9)
miRTarBase	curated functional miRNA–target interactions — the experimental core
TargetScan	sequence (context++) target predictions
ENCORI	Argonaute-crosslink / degradome occupancy
miRGeneDB	guide-versus-passenger arm curation
OmniPath	directed, signed gene–gene interactions for program topology (§5)
COSMIC Cancer Gene Census / OncoKB	232 curated drivers (114 oncogene, 106 tumour-suppressor, 12 dual) for gene-role
DepMap (CRISPR essentiality, 96 breast cell lines)	continuous, all-gene dependency weight (§5)
Lambert 2018 TF census (1,639 human transcription factors)	master-regulator identity within programs (§5)

All sample counts are approximate and refer to the samples passing each analysis's join and quality filters.

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2. The hierarchy of resolution

The same pressure object is modeled at five nested resolutions, each contained in the one before it:

Universe > Program > Target-set > Gene > Edge

— that is, the whole program universe (all 50 programs, 4,384 genes of which 1,410 are under pressure, 721 miRNA arms, 5,108 edges); one program; the genes a single miRNA arm or seed family represses; one gene with its full crowd of regulators; and one miRNA-to-gene interaction. Zooming in localizes a signal, zooming out contextualizes it; each level is one row of the table below.

Resolution	Unit	Question	Pressure quantity	Realization quantity
Edge	one interaction	Does this miRNA repress this gene?	contribution $c(m, g, s)$	per-edge partial correlation
Gene	a gene + its regulators	How much pressure lands here; who carries it?	$\sum_m c$; gene-side share	gene-level net-repression
Target-set	one arm's / family's targets	Does the arm repress its whole module together?	per-arm mass / share	module composite correlation
Program	one program's genes	Net effect on the program's *output*	architecture score	fraction of program coupled
Universe	all programs	Cross-program structure, subtype programs	unsupervised factors	fraction of programs coupled

Two features make this more than "sum the small things":

- **Scope recomputation.** Every mean, standard deviation, median, and softmax that feeds pressure is recomputed *within the samples of the current scope* (whole cohort, one subtype, or one tissue state). A subtype table is therefore not the cohort numbers re-sliced — relative abundances are re-derived inside the subtype.
- **Aggregation carries meaning.** A gene's pressure is the **sum** over its regulators; a program's effect is a **topology-weighted, sign-aware propagation** over its genes, not a sum. The roll-up rule is part of the model.

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3. The interaction universe

The edges are the skeleton over which pressure is built. The universe is **primarily experimental, with a controlled computational extension.**

3.1 Experimental core, computational extension

Layer	Source	Contribution	Role
Experimental core	**miRTarBase** — curated functional miRNA–target interactions (reporter assays, protein blots, qPCR, perturbation, crosslinking)	literature-validated edges	the **spine** — only these enter the default pressure budget
Sequence prediction	**TargetScan** — seed-match + conservation (context++) scores	binding-site strength, study-count-independent	extension, discovery screens, hybrid sensitivity
Crosslink depth	**ENCORI** — Argonaute-crosslink / degradome occupancy	physical-binding depth	a depth boost on the subset of pairs with such data

The default spine is the curated, experimentally supported set (5,108 edges over 721 expressible miRNA arms and 1,410 target genes). Sequence-only ("orphan") predictions do **not** enter the default budget — they are screened separately for discovery (§9) so that prediction noise can never inflate a headline.

3.2 The high-evidence gate

An interaction qualifies for the spine if it has functional support from a *direct* assay — at least one functional study, including at least one reporter or protein assay. This gate is a binary membership filter (which edges are credible enough to model); the weight below is the continuous strength of those that pass.

3.3 Interaction weighting

The weight is an **evidence numerator** divided by a **promiscuity denominator**.

Evidence numerator The numerator does not use a raw study count, which inflates with publication volume and treats a proximity hit like a functional proof. Instead it is a quality-weighted, log-damped sum over assay classes. Writing $\ell(n) = \log(1+n)$ for the study count of one assay class,

$$\mathrm{ev}(m,g) \;=\; 3.0\,\ell(n_{\text{reporter}}) + 2.5\,\ell(n_{\text{protein}}) + 1.5\,\ell(n_{\text{rna}}) + 1.5\,\ell(n_{\text{pert}}) + 0.5\,\ell(n_{\text{bind}}) \;+\; 0.3 \sum_{\text{weak classes}} \ell(\cdot).$$

Three rationales are built in:

- **Assay directness** — a reporter assay that *demonstrates* repression earns six times a crosslink hit that only shows *proximity*. This single correction stops physical-occupancy data from masquerading as functional proof.
- **Log damping within each class** — the tenth confirming paper adds little over the third; $\ell(\cdot)$ gives diminishing returns and defuses publication-count bias.
- **A weak-evidence discount** (0.3) — curator-flagged weak evidence still counts, at thirty percent.

These class weights are a *transparent prior, not a fit*. Perturbing all of them — random $\pm$25% jitter and a systematic $\pm$50% on each in turn — leaves the realized-coupling headline essentially unchanged (the count of strong-negative genes stays within ~10% of baseline, coefficient of variation ~0.06), so no result rests on the specific numbers.

Where physical-binding depth is available it is added (with overall weight 0.5):

$$\begin{aligned}\mathrm{enc}(m,g) &= 2\,\ell(\text{crosslink depth}) + \ell(\text{degradome depth}) \\ &\qquad + 0.5\,\mathbb{1}[\text{three predictors agree}] + 0.25\,\ell(\text{pan-cancer depth}), \\ \mathrm{ev}_{\mathrm{eff}}(m,g) &= \mathrm{ev}(m,g) + 0.5\,\mathrm{enc}(m,g).\end{aligned}$$

Promiscuity denominator A miRNA that targets everything must not dominate any one gene merely by being promiscuous. So the base strength $\ell(\mathrm{ev}_{\mathrm{eff}})$ is divided by the arm's total evidence mass:

$$D(m) = \ell\Big(\sum_{g'} \ell(\mathrm{ev}(m,g'))\Big), \qquad w(m,g) = \frac{\ell\big(\mathrm{ev}_{\mathrm{eff}}(m,g)\big)}{\max\big(D(m),\varepsilon\big)}.$$

What the denominator does, and does not, affect — measured, not asserted. Because $D(m)$ is a single per-arm scalar, it is invisible to two of the three things one might think it tunes, and we verified both directly rather than claim a coupling benefit it does not have. (i) *Realized coupling* is essentially unchanged: in patient cross-validation the denominator is coupling-neutral — the unnormalized weighting is in fact marginally ahead on out-of-sample coupling, and the spine's choice sits within ~0.001 of it (and the candidate ranking reproduces perfectly across held-out folds, so nothing here is overfit). (ii) *Specificity* is mathematically invariant to it: $\mathrm{spec}(m,g)$ is a within-arm ratio in which $D(m)$ cancels exactly (verified: rank-correlation 1.0). Where the denominator *does* act is the one place it should — **cross-arm attribution**: it changes which arm is credited as a gene's dominant regulator for about a tenth of genes and measurably de-concentrates the hub structure (a handful of promiscuous high-evidence arms otherwise dominate disproportionately many genes; the normalization spreads that dominance from ~43 to ~56 arms). So the denominator earns its place as **promiscuity control on the attribution axis**, not as a coupling lever — and penalizing by total *credible evidence mass* rather than raw target count is a deliberate middle (a pure degree penalty de-concentrates slightly more), not a uniquely optimal choice.

3.4 The two directions of many-to-many

The two directions of the graph are handled by two different mechanisms; conflating them is a common error.

- **One miRNA, many genes** is controlled by the denominator $D(m)$ above and by *specificity* (§5.4), which measures how concentrated an arm's budget is on a single gene (focused target versus promiscuous hub).
- **Many miRNAs, one gene** is handled by gene-level aggregation ($\sum_m c$) and by *gene-side share* — the fraction of a gene's total incoming pressure that one arm carries, with a rank (rank 1 = the gene's dominant regulator). A heavily targeted gene may have dozens of regulators; gene-side share lets us name the dominant one while recognizing that the same edge is a trivial slice of that miRNA's own portfolio. The arm-side and gene-side shares answer opposite questions and both are kept.

A gene that belongs to several programs generates one edge row per program for program-level views; cohort-level counts deduplicate per interaction so a multi-program gene is not over-counted.

3.5 The pan-context caveat and non-circular refinements

The curated evidence carries no tissue field — an interaction is "high-evidence" if it was validated in *any* cell line. In this cohort only about an eighth of high-evidence edges have any breast-specific validating paper. Rather than hide this, two refinements quantify it, and both are kept strictly separate from the pressure budget so they remain non-circular:

- a **tissue re-ranking** that re-orders a gene's regulators by adding breast-literature weight, recovering textbook tissue-specific assignments; and

- a **multi-axis prior** that composes evidence with sequence strength, tissue literature, *negative* (contradicting) evidence, and guide-versus-passenger arm status, each as a separate, visible term.

Both are **audit and nomination overlays, never folded into the default budget**, for three reasons. First, *non-circularity*: the budget is later tested against patient expression, so any term we might also want to validate against that same expression must stay out of the budget, or the test becomes circular. Second, *unvalidated utility*: a tissue re-ranking can only be shown to be an *improvement* against an independent breast-specific miRNA dataset, and no suitable one exists (§10), so we do not trust it enough to redefine the budget. Third, *they do not move the headline*: the refinements reorder which specific arm is credited for a given gene but leave the hub- and family-level structure unchanged — so their value is per-target nomination, not a change to the aggregate field. Keeping them as overlays lets each be inspected and tested on its own rather than silently baked in.

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4. Expression and pressure

4.1 The dynamic ingredient

All sample- and state-to-state variation enters through expression. Let $x_{m,s} = \log_2(\mathrm{RPM}+1)$ be arm m's abundance in sample s, with in-scope mean μ_m, standard deviation σ_m, and median $\tilde{x}_m$, and let h_m be the arm's median in healthy tissue. Three building blocks are combined into modes:

$$\underbrace{z_{m,s} = \frac{x_{m,s}-\mu_m}{\sigma_m}}_{\text{within-tumour dynamics}} \qquad \underbrace{\mathrm{sm}_{m,s} = \frac{e^{\,x_{m,s}-\tilde{x}_m}}{\sum_{m' \in R(g)} e^{\,x_{m',s}-\tilde{x}_{m'}}}}_{\text{relative share among a gene's regulators}} \qquad \underbrace{\mathrm{dev}_{m,s} = x_{m,s} - h_m}_{\text{deviation from healthy}}$$

The standardized score z measures how far an arm sits from *its own* typical level — so a stably high arm sits at its own mean (z near zero) and is invisible under z (intended for coupling, where dynamics are the signal; wrong for attribution). The softmax sm is the fraction of a gene's regulator "abundance budget" an arm holds, and is where the *crowd of regulators* enters. The healthy deviation dev has no standard-deviation division, so a uniformly tumour-elevated oncomiR is not gated to zero — this drives the acquired axis (§8). An absolute-level anchor $\max(x_{m,s}, 0)$ keeps the abundance scale in the score.

These combine into three **pressure-construction modes**. The names denote *which downstream analysis each construction feeds* — not a transform applied edge-by-edge inside a correlation:

spine (z):	$\mathrm{expr}(m,s) = \mathrm{sm}_{m,s} \cdot z_{m,s} \cdot \max(x_{m,s}, 0)$	(share × dynamics × level — feeds aggrega
attribution (no-z):	$\mathrm{expr}(m,s) = \mathrm{sm}_{m,s} \cdot \max(x_{m,s}, 0)$	(nothing zeroed for being stable — feeds sha
acquired:	$\mathrm{expr}(m,s) = \mathrm{sm}_{m,s} \cdot \mathrm{dev}_{m,s} \cdot \max(x_{m,s}, 0)$	(centred on healthy, not tumour)

Using the wrong construction silently corrupts the answer: dynamics for the coupling-facing spine, no-dynamics for attribution (who carries the steady-state pressure), healthy-centred for acquisition. **Important:** these build the *aggregate* pressure (over a gene or program). A single-edge or single-module coupling does **not** apply this transform — it uses the arm's raw abundance directly; §6.3 makes the predictor-by-resolution explicit.

4.2 The master formula

Everything downstream is a piece of, or a summary of, this:

$$c(m,g,s) = \mathrm{expr}(m,s)\cdot w(m,g), \qquad \mathrm{pressure}(g,s) = \sum_{m\in R(g)} c(m,g,s).$$

The contribution $c(m,g,s)$ is the **edge-resolution pressure atom**; a gene's pressure is its sum over the gene's regulators $R(g)$. The default aggregation is the sum (a gene with more, stronger regulators genuinely carries more pressure); the mean is a sensitivity variant.

An optional **availability gate** scales pressure by per-sample RISC capacity, mapped through a bounded sigmoid into $[g_{\min}, 1]$:

$$\mathrm{gate}(s) = g_{\min} + (1-g_{\min})\,\sigma\big(k\,(\kappa(s)-c_0)\big).$$

Repression is not free: a miRNA acts only once it is **loaded into an Argonaute** and the loaded complex **engages a TNRC6/GW182 effector** that recruits the degradation machinery, and both steps draw on a finite pool every miRNA competes for — which is why a per-sample, miRNA-independent capacity term is the right shape for a global gate. Capacity therefore has two both-required components, each a standardized abundance: a **loadable-core** term $\mathrm{ago}(s)$ — an abundance-weighted mean over AGO1–4 that up-weights AGO2, the predominant and only slicer-competent Argonaute — and an **effector** term $\mathrm{eff}(s)$ over TNRC6A/B/C. Because either resource can be rate-limiting, capacity is their **co-limiting minimum** (Liebig's law of the minimum, not a sum — abundant AGO must not paper over absent effector):

$$\kappa(s) = \min\big(\mathrm{ago}(s),\, \mathrm{eff}(s)\big).$$

Here $\sigma(\cdot)$ is the logistic sigmoid, $g_{\min}\in(0,1)$ the floor (the gate never falls below it, so the layer can only attenuate, never erase — even a low-capacity tumour keeps basal repression), c_0 the capacity midpoint, and k the slope. The AGO weighting reflects abundance dominance, not catalysis (slicing matters mainly for the rare extensively-complementary site). It is reported as a sensitivity layer, not a mechanism — AGO/TNRC6 *mRNA* is an imperfect readout of functional RISC capacity, so the bounded form is deliberately conservative, and empirically it rescales rather than reorders programs.

4.3 Acquired and lost pressure, at every resolution

Acquired pressure is what the tumour gained over the healthy state; lost pressure is what it shed. It is defined identically at each resolution by differencing the same contribution across states (the cross-state comparison uses the no-dynamics mode, since z is standardized *within* a state and is not comparable across states):

$$\begin{aligned}
\text{edge:}\quad & \mathrm{gain}(m,g) = c_{\text{tumour}}(m,g) - c_{\text{healthy}}(m,g),\\
\text{gene:}\quad & \mathrm{acquired}(g) = \mathrm{pressure}_{\text{tumour}}(g) - \mathrm{pressure}_{\text{healthy}}(g),\\
\text{program:}\quad & \Pi_{\text{acq}}(m,\mathcal{H}) = \mathrm{prior}(\mathcal{H})\cdot\sum_g \big[-\max(c_{\text{tumour}} - c_{\text{healthy}}, 0)\cdot \mathrm{infl}(g)\big].
\end{aligned}$$

(The program-level form previews the architecture layer of §5: $\mathrm{infl}(g)$ is the gene's net downstream influence *within the program* — how much repressing g changes the program's output, defined in §5.1 — and $\mathrm{prior}(\mathcal{H})$ is the program's tumour-direction polarity, §5.2.)

Two healthy anchors are complementary: a true-healthy reference (large baseline jumps, the canonical oncomiR program) and a normal-adjacent reference on the *same* measurement platform (no cross-platform artifact, ranking the final malignant step). Section 8 gives the trajectory machinery in full.

4.4 Share and specificity

Share decomposes a gene's pressure among its regulators. Writing the over-bar for the mean over samples and a dot for the sample index it runs over — so $\overline{|c(m,\cdot)|}$ is the across-sample mean of an edge's absolute pressure, and $\sum_{m'}$ runs over the gene's regulators —

$$\text{abs-share}(m) = \frac{\overline{|c(m,\cdot)|}}{\sum_{m'} \overline{|c(m',\cdot)|}} \quad \text{(ranking)}, \qquad \text{signed-share}(m) = \frac{\overline{c(m,\cdot)}}{\sum_{m'} \overline{c(m',\cdot)}} \quad \text{(coherence diagnostic)}.$$

When the signed share diverges from the magnitude share, regulators are *cancelling*. Under the no-dynamics attribution mode every contribution is non-negative, so the two coincide — the reason attribution is computed there.

Specificity is the fraction of an arm's *total* target-mass — summed over all the genes $T(m)$ it regulates — that lands on one gene:

$$\text{spec}(m,g) = \frac{\overline{|c(m,g,\cdot)|}}{\sum_{g' \in T(m)} \overline{|c(m,g',\cdot)|}}.$$

The reference point is the even-spread baseline $1/|T(m)|$: an arm with dozens to hundreds of targets that spread its budget evenly would put well under one percent on each. So a value of a *few* percent already marks a gene that commands a disproportionately large, focused share of the arm's budget — several-fold above the even-spread floor — and hence a cleaner single-gene readout (here $\gtrsim 0.05$). A value at or below that floor ($\lesssim 0.01$) marks a gene the arm regulates only as one of a promiscuous crowd. (It is a *relative concentration*, not an absolute majority — even a focused arm rarely puts most of its budget on a single gene.)

> **Seed-family non-identifiability — a caveat on *every* single-arm number.** Share, gene-side rank, and specificity all credit a *named arm*, but paralogues that share a seed (for example the miR-29a/b/c or miR-302/372/373/520 families) target the same sites and co-vary in expression, so a per-arm attribution cannot say *which* family member actually carries the pressure — the credited arm may be standing in for its family. This is not a peripheral worry: across the cohort, within-seed-family arm pairs co-vary about twice as strongly as random pairs and the largest multi-arm hubs are 20–37% redundant (one effective regulator masquerading as several). We therefore read share/specificity at the **seed-family** level as the safe unit and treat the specific within-family arm assignment as a nomination, not a claim; the realized-coupling inference makes the same grouping explicit (the seed family is the unit for both the dependence-aware FDR and the permutation null — §6, §9.3). Singleton-seed arms are unaffected.

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5. Architecture: modeling a set of genes

A program is not a bag of independent genes. A miRNA that represses an upstream activator changes the whole downstream program; one that represses a tumour-suppressor is pro-tumour regardless of network position. The architecture layer overlays three orthogonal lenses on otherwise-flat pressure.

5.1 Topology — network position

Build the program's regulatory wiring from **OmniPath** — a curated meta-database of directed, signed (activating / inhibiting) molecular interactions — induced on the program's member genes. Each ordered gene pair gets a **calibrated continuous sign**, so a conflicting (dual-mode) pair contributes a graded value instead of being flattened:

$$\mathrm{ns}(s \to t) = \frac{n_+ - n_-}{n_+ + n_-} \in [-1, +1].$$

A finite-horizon signed-path propagation then gives each gene its net downstream influence:

$$T = \sum_{k=1}^{H} (\alpha S)^k, \qquad \mathrm{infl}(g) = 1 + \sum_j T_{g,j} \qquad (\alpha = 0.25,\ H = 6),$$

where S is the signed adjacency. A central activator scores positive, an inhibitor hub negative. Because a miRNA *represses* its target, the effect on program output flips with the node's influence:

$$\Delta_{\text{prog}}(m) = \sum_g \big[- c_{\text{tumour}}(m, g) \cdot \mathrm{infl}(g) \big].$$

This is the only lens that captures *indirect*, network-mediated effects — everything else is local.

5.2 Program-polarity prior — pro- or anti-tumour

A coarse, curated polarity per program ($+1$ pro-tumour engines such as EMT, hypoxia, glycolysis, E2F, G2M, MYC; -1 tumour-suppressive or anti-tumour-immune programs such as apoptosis, p53, interferon, inflammatory, immune-rejection; 0 ambiguous, kept directional but unpolarized):

$$\Pi(m, \mathcal{H}) = \mathrm{prior}(\mathcal{H}) \cdot \Delta_{\text{prog}}(m).$$

A positive value pushes toward malignancy — either by activating a $+1$ engine or by **releasing a brake** on a -1 program. A clean mechanistic result follows directly: because miRNAs repress, they *damage* the proliferation engines (negative pro-tumour on E2F, G2M, EMT) and turn malignant mainly by **brake-release on the suppressive programs** (for example, the canonical oncomiR releasing the p53 and apoptosis brakes, and another releasing immune-evasion). Every program is, on net, *damaged* by the miRNA layer; only the targeted sub-pattern is oncogenic.

5.3 Gene-role and malignancy — driver-level direction

Topology has a blind spot for canonical oncomiRs, because they mostly damage the engines they broadly hit, and because some key targets act through metabolites rather than gene nodes. A curated **gene-role** layer (high-confidence cancer drivers, each labelled oncogene, tumour-suppressor, or dual) provides a per-gene malignancy sign $s_{\text{mal}}(g) \in \{-1, 0, +1\}$:

$$\mathrm{Mal}(m) = \sum_g \big[- s_{\text{mal}}(g) \cdot c_{\text{tumour}}(m, g) \big], \qquad \mathrm{Mal}_{\text{arch}}(m) = \sum_g \big[- s_{\text{mal}}(g) \cdot c_{\text{tumour}}(m, g) \cdot w_{\text{arch}}(g) \big],$$

where the non-negative centrality weight w_{arch} scales magnitude without re-flipping the sign. This recovers the textbook oncomiR roster that topology missed — external validation that the layer captures real driver biology.

> Topology and gene-role are kept side by side, not merged: they answer different questions — "what the wiring says" versus "what the driver annotation says" — and the canonical oncomiR is exactly the case where both must be read. For broad, abundant arms, the gene-role, coupling, and acquired axes are the reliable readouts, not the topology sign.

A continuous, **all-gene** complement uses functional-genomic dependency (genome-wide CRISPR essentiality in breast cell lines): genes the tumour depends on (strongly essential) versus tumour-suppressor-like genes,

$$\mathrm{Mal}_{\mathrm{cont}}(m) = \sum_{g} w_{\mathrm{dep}}(g) \cdot c_{\mathrm{tumour}}(m, g),$$

which extends coverage from a few hundred curated drivers to nearly all genes and shows the miRNA layer landing hardest on proliferation dependencies. A transcription-factor census adds a master-regulator identity flag on top of pure topology centrality.

5.4 Roll-ups

Each architecture score rolls up across the universe (per arm across programs, per program across arms, by seed family) and runs in an **acquired-weighted** parallel that isolates the tumour-acquired pro-tumour pressure from the constitutive component. The universe-level unsupervised structure — pressure programs and their subtype divergence — is described in §9.2.

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6. Coupling: realized pressure and confounding

Pressure is potential. **Coupling** asks whether repression is actually observed: does a tumour with more of the miRNA carry less of the target, after the obvious confounders are removed?

6.1 The core idea

Realized repression should appear as a **negative correlation** between the miRNA *dose* and the target's expression. The dose is the arm's **raw abundance** at single-regulator resolution (one edge, or one arm's target module) and the **aggregate pressure** — the weighted, expression-transformed sum over regulators (§4) — where many regulators converge on one gene or program (the only way to fold them into a single predictor). A naive correlation, however, is dominated by confounders — a more cellular, more aneuploid, more proliferative tumour drags everything with it. We therefore use a **partial Spearman correlation**: regress the confounders out of *both* variables, then correlate the residual ranks. A negative partial correlation that survives is realized repression. Rank correlation (not linear) is used because expression is heavy-tailed and the expected relationship is monotone, not strictly linear.

6.2 The confounders

Each canonical confounder is included for a specific mechanistic reason.

Confounder	What it is	Why it confounds
Tumour purity	fraction of malignant cells	stroma and immune infiltrate carry their own miRNAs and mRNAs, so a low-purity tumour looks “high miRNA / low target” compositionally
Genomic instability	homologous-recombination deficiency / scar burden	instability co-drives both arms; the most unstable subtype over-expresses many loci
Target copy number	the target gene's own copy number	an amplified target's mRNA rises for reasons unrelated to the miRNA, masking or faking repression
Proliferation	a cell-cycle expression signature	the cell cycle drags both the proliferation-miRNA clusters and their targets and must be removed
Composition (immune/stromal/epithelial)	deconvolved cell-content scores	the most complete composition control; the epithelial axis specifically de-confounds lineage co-expression

Confounder	What it is	Why it confounds
Sequencing batch	technical run/plate identity (the small-RNA and mRNA assay plates; per-state batch for the healthy and normal-adjacent references; the protein-assay plex for the proteome cohort)	a shared technical batch co-varies features on *both* arms, so it can fabricate **or** mask coupling; it is removed as **covariate dummies**, not by batch-effect *subtraction* (which over-corrects an unbalanced design and can manufacture the very feature–feature correlations that are the readout)

Three subtleties make the proliferation handling careful rather than reflexive:

- **Three proliferation proxies** of differing cell-cycle-gene dependence are used in turn, to test whether repression survives proliferation adjustment *without over-correcting through a shared pathway* — the “we just removed the signal” objection.
- **Proliferation is a *suppressor* confound, not the source of the signal.** Within the most proliferative subtype it associates positively with *both* pressure and target expression, so adjusting for it makes the negative coupling *more* negative — it was masking, not faking. The coupling therefore strengthens under proliferation adjustment, a robustness result rather than a fragility.
- **A covariate ladder, not a single number.** Coupling is reported across a nested sequence of covariate sets (none → purity + instability → + target copy number → + proliferation), **with sequencing-batch dummies appended to every adjusted rung** (the raw rung is the only batch-free one), so a reader sees exactly what each adjustment costs. The headline significance flag requires a negative partial correlation with a false-discovery rate below five percent; a parallel flag requires survival of the proliferation-adjusted ladder. A variability gate (low standard deviation *and* low interquartile range) marks flat arms whose null coupling should not be over-read.

Multiplicity, dependence, and an empirical null — at every rung. Two precautions guard the significance calls, and both are applied at *each* resolution below, not only the headline one. First, the false-discovery control is made **dependence-aware**: because seed-family paralogues are non-independent (within-family arm pairs co-vary about twice as strongly as random pairs), plain Benjamini–Hochberg over edges would be anti-conservative, so we add a Benjamini–Yekutieli correction (valid under arbitrary dependence) and, where the test unit is a miRNA, collapse each seed family to a single test (Simes) so a multi-paralogue hub spends one slice of the error budget rather than several — the de-duplicated discovery count is reported in *families*, not edges. Second, every reported coupling is re-tested against an **empirical permutation null** (Freedman–Lane residual permutation, one shared sample permutation per draw so the family correlation structure is preserved), so the significance does not rest on the asymptotic partial-correlation p alone; the same joint null yields a count statistic for the universe-level “how many programs couple” claim. These run at edge, gene, target-set, program, and universe resolution alike.

6.3 Coupling at every resolution

The same partial-correlation machinery is applied at each rung of the hierarchy, against a different predictor and target.

Resolution	Predictor	Target	The coupling object
Edge	arm abundance	target expression	per-edge partial correlation, per tissue state; a three-letter trajectory pattern across states
Gene	the gene’s aggregate incoming pressure	the gene’s own expression	whether the *gene* is net-repressed
Target-set	arm abundance	the mean standardized expression over the arm’s targets (the *module composite*)	whether the arm’s whole module moves down together
Family	a seed-family aggregate predictor	target	family-as-unit coupling

Resolution	Predictor	Target	The coupling object
Program	per-program pressure	per-program mean target expression	the count of programs that couple
Universe	all programs	—	the fraction of programs and edges that couple, and cross-cohort concordance

The **module composite** is the strongest confounder-independent signal: standardize each target within a state, average across the arm's targets, then take one partial correlation of the arm's abundance against that composite. A strongly negative value means the *coordinated* module is repressed — something no single-edge correlation can see. A role-stratified version (tumour-suppressor versus oncogene targets) is an interpretive overlay, deliberately not folded into one number, since whether repression is pro- or anti-tumour is the architecture layer's job (§5).

Why the predictor switches with resolution — and what actually differs. The edge and aggregate predictors are *not* a reparametrization of one another, but most of the difference is inert. Of pressure's pieces, the static evidence weight is a sample-constant scalar (invisible to a rank test) and the within-arm z is an affine transform of abundance (rank-identical for a single arm), so neither changes a single-edge coupling; the **softmax share** — which pulls in the co-regulators — is the one piece that does. A head-to-head check (the same confounder-adjusted partial Spearman applied under every predictor) confirms it:

- **Edge level:** the within-arm z reproduces raw abundance (sign-concordance 0.96, recovering 94% of its FDR-significant edges), while the share *alone* recovers only ~49% and flips its median positive. So **raw abundance is the most sensitive single-edge predictor** — which is exactly why edge and module coupling use it.
- **Gene level:** the share alone is degenerate (per-gene shares sum to about 1 in every sample, so the predictor is near-constant), but the *full* aggregate pressure (share $\times$ dynamics $\times$ level) recovers ~89% of a naive abundance-sum's hits **and finds ~2–3% more, with a deeper median** — so the pressure construction is a genuinely better way to fold many regulators into one dose than summing raw abundances.

Net: pressure differs from abundance **only** through the softmax share, and that term *weakens* a single edge but *strengthens* the gene aggregate — which is why the predictor is abundance at edge/module resolution and aggregate pressure at gene/program resolution.

6.4 The potential $\times$ realized join (pressure $\times$ coupling)

The purpose of measuring both potential and realization is to **join** them. In this join *pressure* is the **potential** axis (at edge level its trajectory tracks the miRNA's abundance, since the edge weight is static across states) and *coupling* is the **realized** partial correlation — so the join is potential $\times$ realized, **not** a correlation of one against the other. Each edge carries a pressure (potential) trajectory — does the miRNA gain or lose abundance across states — and a coupling (realized) trajectory — the per-state significance pattern. Their join is the headline class:

- **acquired-and-realized** — pressure rises *and* coupling turns repressive (a newly switched-on repressor);
- **field-established** — already repressive in normal-adjacent tissue, retained in tumour;
- **acquired-unrealized** — pressure rises but coupling never realizes;
- **released brake** — repressive in healthy tissue, gone in tumour;
- **constitutive** — repressive in both healthy and tumour.

This join lets us state, as a *realization* claim immune to the pressure-side caveats, that healthy breast actively holds certain oncogenic and invasion targets under miRNA repression and the tumour releases the brake. Whether such a loss reflects a genuine loss of repression is then tested by conditioning on copy number, promoter methylation, and chromatin accessibility one at a time (§9).

7. Protein realization: beyond the messenger-RNA layer

miRNA repression ultimately acts on the **protein** product. The messenger-RNA-level coupling sees destabilization but cannot see a pure translational block, so realization onto protein is tested directly, three ways, using mass-spectrometry proteomes. For a target with standardized mRNA and protein levels,

$$\begin{aligned} &\textbf{direct protein:} && \rho\big(\text{pressure},\ \text{protein}\big) && \text{expected negative} \\ &\textbf{messenger-to-protein gap:} && \rho\big(\text{pressure},\ \text{mRNA} - \text{protein}\big) && \text{expected positive} \\ &\textbf{protein beyond mRNA:} && \rho\big(\text{pressure},\ \text{protein} \mid \text{mRNA}\big) && \text{expected negative} \end{aligned}$$

The third quantity — protein residualized on its *own* messenger RNA — is the key idea: it removes everything the **measured** mRNA already explains, so a surviving negative correlation is a protein-level effect that the bulk steady-state mRNA does **not** capture. This is the distinction between reading protein *together with* the mRNA coupling (direct) and *after subtracting* it (residual).

It is tempting to call that residual "translational repression," and it is consistent with a translational block — but it is **not** exclusively that, so we deliberately label it *beyond-(measured)-mRNA* rather than *translational*. At least four non-translational routes produce the same residual: (i) **destabilization that bulk RNA-seq misses** — miRNA-driven deadenylation / poly(A) shortening lowers a message's output before its total count drops, and poly(A)-selected libraries register this differently from total-RNA libraries; (ii) **3'-UTR isoform / alternative-polyadenylation effects**, where gene-level mRNA quantification averages over isoforms that differ in whether the miRNA site is even present; (iii) **protein-versus-mRNA turnover kinetics** — protein level is a time-integral of past mRNA, so a transient or recently-onset mRNA dip can surface in protein while being smoothed out of a single steady-state mRNA snapshot; and (iv) **indirect routes**, where the miRNA represses an intermediate that changes the target's protein without changing the target's own mRNA. Platform noise between sequencing and mass spectrometry adds further slack. The residual is therefore best read as *regulation the mRNA snapshot cannot see*, of which translational repression is one — biologically expected — component.

The headline findings: the spine validates at the protein layer in an *independent* patient cohort (dozens of genes show significant negative pressure-to-protein coupling, led by epithelial-keratin, hypoxia, and matrix targets — KRT8, DDAH1, BMP1 — *not* a canonical EMT transcription factor, ZEB1/2 being absent from the protein-layer drivers); the repression is **predominantly mRNA-mediated** (residualizing on mRNA collapses most of it, leaving a small beyond-mRNA residual — consistent with canonical mammalian miRNA biology, in which destabilization dominates and a smaller component acts beyond the steady-state message); and the surviving residuals are **known, subtype-structured edges**, including a protein-level recovery of the central proliferation-cluster-to-tumour-suppressor hub. A decoy control confirms the residual couplings are **target-specific** — each driver arm anti-correlates with its cognate protein and not with the proteins it does not target — and they survive batch adjustment.

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8. The healthy → normal-adjacent → tumour trajectory

How a miRNA's pressure *moves* across tissue states is itself a finding, so the acquired concept gets dedicated machinery across three states: true-healthy tissue, normal-adjacent tissue (which carries a pre-malignant field effect), and tumour.

8.1 The healthy anchor

The healthy reference is **GTEx breast** (346 normal-tissue samples), quantile-normalized onto the tumour-cohort scale and joined by stable molecular accession. Because the GTEx pipeline collapses a subset of canonical arms to zero, two imputation layers repair coverage — a validated rank-transfer from an **independent healthy miRNA atlas (DIANA-miTED)** (primary) and a seed-family fallback — while arms with no reference anywhere are left explicitly un-imputed and flagged rather than fabricated, since imputing a truly absent arm would erase a real acquired gain. The healthy abundance matrix is repaired *at source* so the acquired axis is free of the collapse artifact, while the realized-coupling layer is left on the raw matrix and so is unaffected. The imputed baseline feeds only the acquired axis, never the coupling — so all headline coupling findings are baseline-independent.

8.2 Two acquired axes

A single rank delta is insufficient, because rank *saturates*: an already-abundant arm cannot gain rank even as its absolute abundance rises several-fold. Two complementary axes are therefore used,

$$\underbrace{d_{HT}(m) = r_{\text{tumour}}(m) - r_{\text{healthy}}(m)}_{\text{rank delta (saturating, cross-platform-robust)}} \qquad \underbrace{\text{log2fc}(m) = \tilde{x}_m^{\text{tumour}} - b_m^{\text{healthy}}}_{\text{magnitude delta (recovers the surge)}},$$

where r is a within-state percentile rank and b the healthy baseline on the common scale. The magnitude axis recovers exactly the strongly up-regulated oncomiRs the rank axis is blind to. The default acquired call is the union: an arm is an acquired gainer if it gains on *either* axis.

8.3 Trajectory classification

The three legs are healthy→normal-adjacent (the field effect), normal-adjacent→tumour (the tumour-specific increment), and healthy→tumour (the acquired headline). On these, each miRNA is classed as stable, progressive gain or loss, tumour-acquired, field-effect, or non-monotonic; each edge inherits a class from the join of the miRNA's movement with its per-edge coupling trajectory (§6.4); and a subtype-specificity index marks whether an acquired arm is pan-subtype or concentrated in one subtype. Because the acquired axis is anchored to the true-healthy state — never the field-affected normal-adjacent tissue — "acquired over healthy" means exactly that.

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9. Validation

Findings are graded by the strength of evidence behind them: **statistically supported** (a significant, confounder-adjusted effect), **descriptive** (a coherent pattern), or **a nomination** (a hypothesis awaiting a perturbation assay). The three validation questions are: did the model recover known biology, did it deepen it, and did it find anything new?

Underpinning all three is an **inference-rigor layer** that guards the usual soft spots. Every coupling claim is re-tested at *each* resolution against an empirical permutation null (so significance does not rest on an asymptotic p alone) and a **dependence-aware** false-discovery control — Benjamini–Yekutieli plus, where the unit is a miRNA, a seed-family collapse, so co-varying paralogues cannot each spend an independent slice of the error budget. The pressure construction itself is validated **out-of-sample**: in patient cross-validation its candidate ranking reproduces on held-out patients, so it is not overfit to the cohort it is tested against, and the weighting constants are stability-tested rather than asserted.

9.1 Recovered — the positive controls

The model reproduces textbook biology *unprompted* — recovered from inputs that never saw the patient expression data — which is evidence that the recovered structure is not an artifact of the construction. The canonical tumour-suppressor-miRNA roster falls out of the architecture anti-tumour leaders, and the canonical oncomiR roster out of the gene-role score. The established miRNA-to-extracellular-matrix, proliferation-cluster-to-tumour-suppressor, and let-7-to-mitotic axes all re-emerge as unsupervised sub-modules — internal validation, since their *absence* would make the decomposition suspect. High-evidence edges couple more strongly than sequence-only predictions in two independent cohorts, and the miRNA-to-target anti-correlation **replicates in an independent patient cohort** (a clear positive concordance, with two-thirds of significant repressive edges keeping a negative sign) — an independent-patient leg the same-patient proteome cannot provide.

9.2 Deepened — the same players, newly resolved

The genuine methodological contribution is **state-resolved** and **within-program** structure that standard enrichment and single-edge analyses cannot see:

- **State-resolved coupling** — the literature asserts a miRNA-target pair and the miRNA's down-regulation, but rarely measures the brake being *engaged in normal tissue and released in tumour*.
- **Within-program sub-modules** — a program resolves into a shared, pan-subtype backbone plus named subtype-specialist sub-modules (for example a basal immune-restriction module, and tumour-suppressor repression localized to a *minority* sub-module), where ordinary enrichment gives a single score per program.
- **Protein-layer confirmation** of the central hub in an independent proteome.
- **An end-to-end copy-number-to-network attribution** showing that the copy-number-driven component of miRNA dosage reproduces a large fraction of the repressive network and survives the full confounder stack.

9.3 Novel and candidate-novel — the discovery surface

The cleanest discovery surface is the **orphan edges**: sequence- and crosslink-predicted interactions absent from functional curation, kept out of the spine precisely so they can serve as an unbiased screen. Of roughly twenty thousand orphan edges screened against the protein layer, a few thousand are testable, several hundred show protein anti-correlation, a couple hundred survive residualization onto mRNA, and the strongest tier — translational *and* replicated in a second cohort — numbers in the dozens, most of them fully uncurated.

Three convergent strengthenings make these more than a prediction list:

- **Method self-validation** — the established miRNA-to-collagen axis emerges *de novo* at uncurated target genes, providing a credibility anchor.
- **Triple-cohort support** — about a hundred genuine orphan edges keep a negative sign across three independent datasets and two molecular layers (messenger RNA, protein, and an independent cohort), far above the chance rate.
- **A composition-robust core** — re-testing the prediction-only tier after adding subtype, immune, stromal, endothelial, and target-copy-number covariates leaves a clean core of about a dozen-and-a-half de-novo nominations (led by a metabolic convergence) while the composition-loaded artifacts drop out.

  These are **wet-lab nominations** — a prioritized queue for crosslinking and reporter assays — not validated edges. Protein anti-correlation is not direct binding, co-expression confounding persists (especially for matrix genes that co-vary with stroma), and seed-family arms cannot be separated — members of one seed family (for example the miR-29 paralogues a, b and c) share the same seed sequence, hence the same predicted target sites, and co-vary in expression, so an anti-correlation cannot say *which* member drives the repression of a shared target; a nominated arm may simply be standing in for its family. The de-novo recovery of the collagen axis is the anchor; the fully uncurated translational hits and the composition-robust core are the discovery surface.

9.4 The validation ladder

Rung	Evidence	Strength
Reproduces known biology	recovers textbook edges unprompted	established
Robustness & inference	permutation null + dependence-aware (BY / seed-family) FDR at every rung; construction validated out-of-sample; constants stability-tested	statistical
Messenger-RNA coupling	significant, confounder-adjusted partial correlation	statistical
State-resolved trajectory	brake-release and acquired-realized classes	descriptive / hypothesis
Protein layer	independent proteome residual	statistical
Independent cohort	replication in new patients	statistical / established

Rung	Evidence	Strength
Wet-lab queue	nominations for perturbation assays	hypothesis

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10. Design principles and honest limits

The stance, as it cashes out: static evidence times dynamic abundance keeps "what is possible" separate from "what this tumour expresses"; a non-circular prior cannot leak into the findings it is tested against — and this is now *checked, not just asserted*, since the pressure construction reproduces its ranking on held-out patients (so it is not overfit to the cohort it is validated on); two measurements (potential and realization) are joined, never conflated; everything is reported adjusted *and* raw, gated *and* ungated; and every claim is labelled by its rung of evidence.

The standing limits, stated plainly:

- **Gene-gene topology cannot see metabolite-mediated regulation** (a target acting through a lipid rather than a gene node) — a true ceiling, partly worked around by the gene-role layer.
- **The mechanism of a lost coupling is not fully pinned.** Copy number and promoter methylation are ruled out by conditioning, chromatin accessibility is largely null with a small minority at growth and cell-cycle loci, and the leading remaining candidate is 3'-untranslated-region shortening and competitive (decoy-RNA / RNA-binding-protein) regulation.
- **Arms with no healthy reference** are a coverage gap, not acquisition — to be resolved by extending healthy-tissue coverage, never by imputing absent arms.
- **The program-polarity prior is coarse** (one polarity per program). The tissue-specific edge re-ranking (§3.5) is kept as an overlay because proving it actually *improves* on the pan-context weighting would require an independent breast-specific miRNA dataset to test the reassigned regulators against — and no suitable one exists (one major breast cohort lacks miRNA data; another's miRNA is re-derived from the same source rather than being independent). The refinement is built and inspectable, but its utility is not yet externally validated.
- **The pressure construction is one of several near-equivalent choices** — validated as non-overfit out-of-sample, but not uniquely optimal: on realized coupling the bare un-normalized weighting is marginally ahead, and the promiscuity denominator is coupling-neutral, earning its place on attribution (which arm is a gene's dominant regulator) rather than on coupling.
- **A bulk-tissue ceiling** — every quantity is an association, not a binding event; cell-composition is controlled but never fully eliminated.

Glossary

Term	Definition
Pressure	potential repression = expression × interaction weight, summed over a gene's regulators
Contribution $c(m,g,s)$	the edge-resolution pressure atom; every other quantity is built from it
Interaction weight $w(m,g)$	static evidence strength divided by a promiscuity denominator
Acquired / lost pressure	tumour pressure minus a healthy anchor, at edge, gene, or program level
Share	a regulator's fraction of a gene's total pressure (arm-side and gene-side)
Specificity	how concentrated an arm's budget is on one gene (focused target versus promiscuous hub)
Coupling	observed, confounder-adjusted anti-correlation = *realized* pressure (predictor = arm abundance at edge/module resolution, aggregate pressure at gene/program)
Module composite	partial correlation of an arm's abundance against the mean standardized expression of its whole target set
Architecture	topology influence + program-polarity prior + gene-role malignancy, applied to a program

Term	Definition
Released brake	an interaction repressive in healthy tissue and lost in tumour
Acquired gainer	an arm that gains pressure over healthy, on a rank or a magnitude axis
Orphan edge	a sequence- or crosslink-predicted interaction absent from functional curation — the discovery screen
Availability gate	a per-sample bounded multiplier from RISC-machinery abundance — a sensitivity layer, not a mechanism